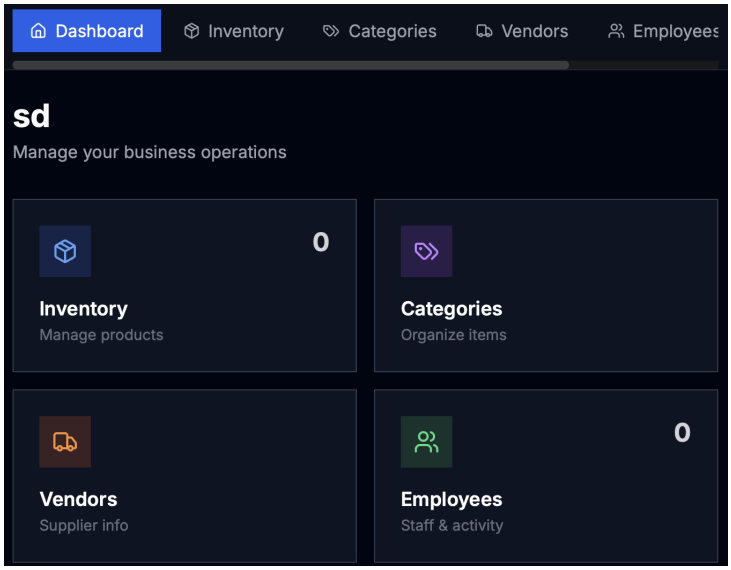
An introduction to what we are looking for:

- A POS system that can be run fully locally using 3-5 terminal computers with their own iteration of the software but drawing inventory/pricing data from the company file stored on a "server computer", which I will build with the appropriate storage and connection capabilities
- The system must be touchscreen friendly in a clean UI, with the particular features that I will highlight below. Some features may take excessive time and development, so I will note which features are priority. Other features that will increase time and cost may be ignored.
- Before getting to design and features, there are a few important capabilities I would like to note:
    - A different interface for the server computer, which will also double as an admin computer for pricing and monitoring purposes, while the terminals will have a purely sales oriented interface
    - We aim to use magellan and NCR scanner scale units. Other units will be generic
    - A way for updates to be added from a flashdrive for example on each terminal and the server if there is ever a feature that needs to be added
    - A way for the terminals to have redundant data saved locally in the event of a disconnect with the server computer, and to save transaction data locally in the event of a disconnect, and to sync with the server when the connection is restored and reconcile data with other terminals
    - A built in capability to create and store (in the inventory) custom barcodes, with the ability for a terminal computer to print them

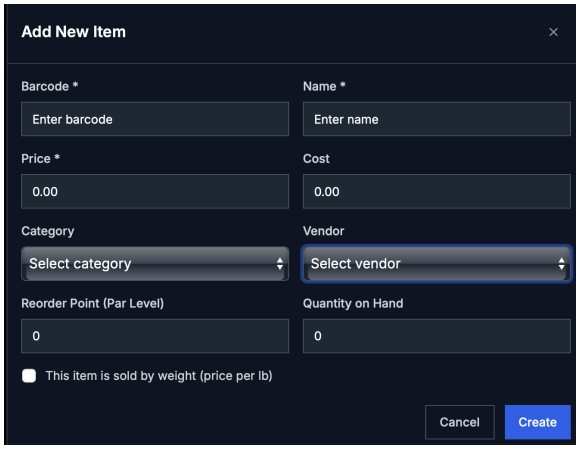
For the actual system design, the installer should ask upon setup whether the computer where the application is to be set up as an admin or terminal computer, and install the according version of the software.

For an admin computer:

The opening screen should allow for the user to select the company file/create one if this is the first setup. After logging in, the following is an example of a preferred layout with neat tiles for various functions.



Beginning with inventory, there should be a search feature to search by name or barcode all of the inventory added. This is also where the user would be able to add items to the inventory. Adding a new item should result in a popup like the image below.



The user can enter the barcode, the name of the item, the price, cost, and category (which will be its own tile that will be described later), the vendor (also its own tile), re order point (also its own tile, but the function is that if the quantity on hand dips to the set quantity, the alert will show up on the reorder page so the user is aware that the item needs to be reordered), and quantity on hand for inventory tracking purposes. As there may be deli items sold by weight, we would like the ability to configure a sold by weight item so that a cashier can be prompted to enter weight instead of qty when such an item is scanned.

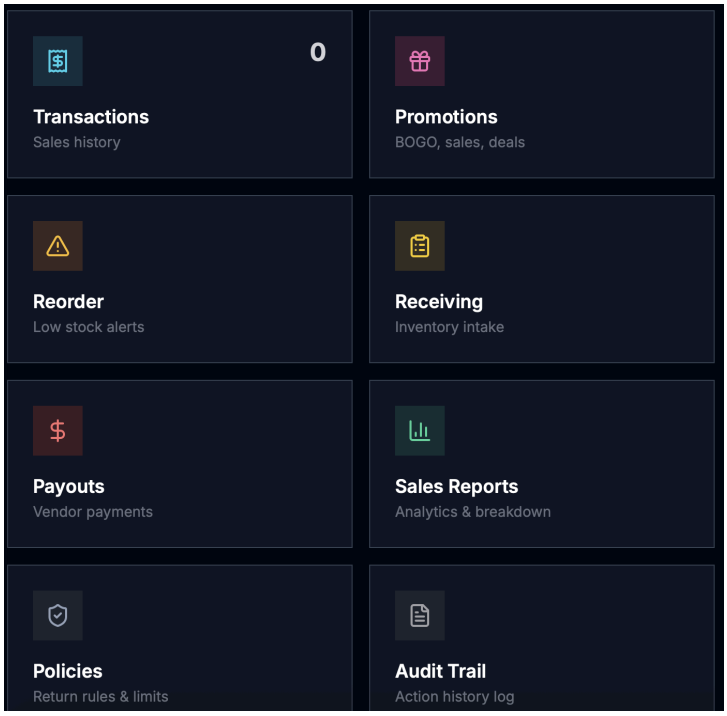
Moving on to the categories tile. This will allow the user to create groups to which items can belong. Adding a new category should prompt the user to enter the name and tax rate if

applicable, as well as whether the category is age restricted. The function is so that the user can view sales by category, apply promotions by category, view taxes collected by category, and apply age restriction so that if an item from such a category is scanned by a sales terminal, a prompt will show up reminding the cashier to verify age.

The vendors tile is quite simple. We just need a function to add our known vendors so that we can attribute them to items as they are added into the inventory. Adding a new vendor should just prompt the user to enter name and contact information.

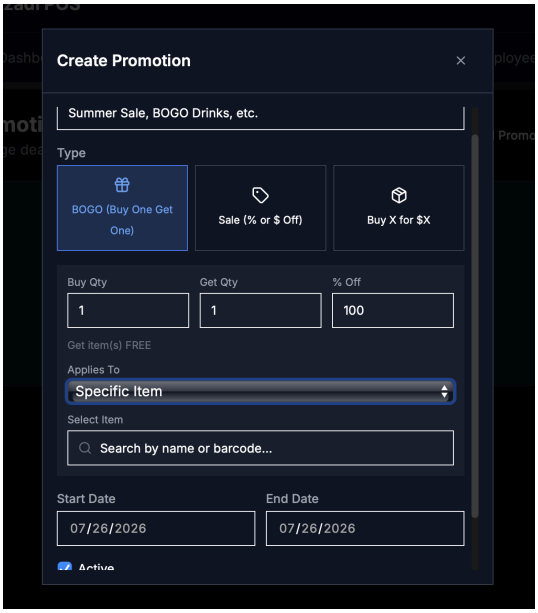
The employees tile is feature rich. We hope for the software to be able to rigorously track every possible metric for monitoring and security purposes. The home page of this tile should show tiles of all employees listed in the database. The add employee button should ask for the name of the employee as well as checkboxes for if they are in sales or if they have manager privileges. Both can be selected. Creating the new employee should generate a barcode associated with a pin (which is printable) that is unique to that employee. This will then be used to scan into the pos terminal. This serves as a way to log in to the terminals as well as tracking metrics by employee. Manager privileges mean that certain actions on the sales terminal, which will be discussed later, will require a managers approval. A barcode with pre configured manager privileges will need to be scanned to approve.

As for actually clicking on an employee, we need to be able to see a broad overview as well as their sales data by shift, with additional filters for categories. We need to flag transactions containing gift cards, refunds, and store credit. Age restricted categories of items sold in a transaction may also be a flag. We also need a feature to compare these metrics with other employees. We should be able to click into the shift of an employee, and see a list of transactions they handled, then click further into them to see what the transaction was.



For the transactions tile, we want to see all sales, refunds, voids, etc. with total amounts as well as a list format for more individualized viewing. We also would like filters including payment method, transaction type, transactions involving credit or gift cards, and date ranges for viewing. The list view of transactions should also allow for deletion in the event of a duplicate or mistake.

For promotions, the below image has been my desired UI for creating and applying them:

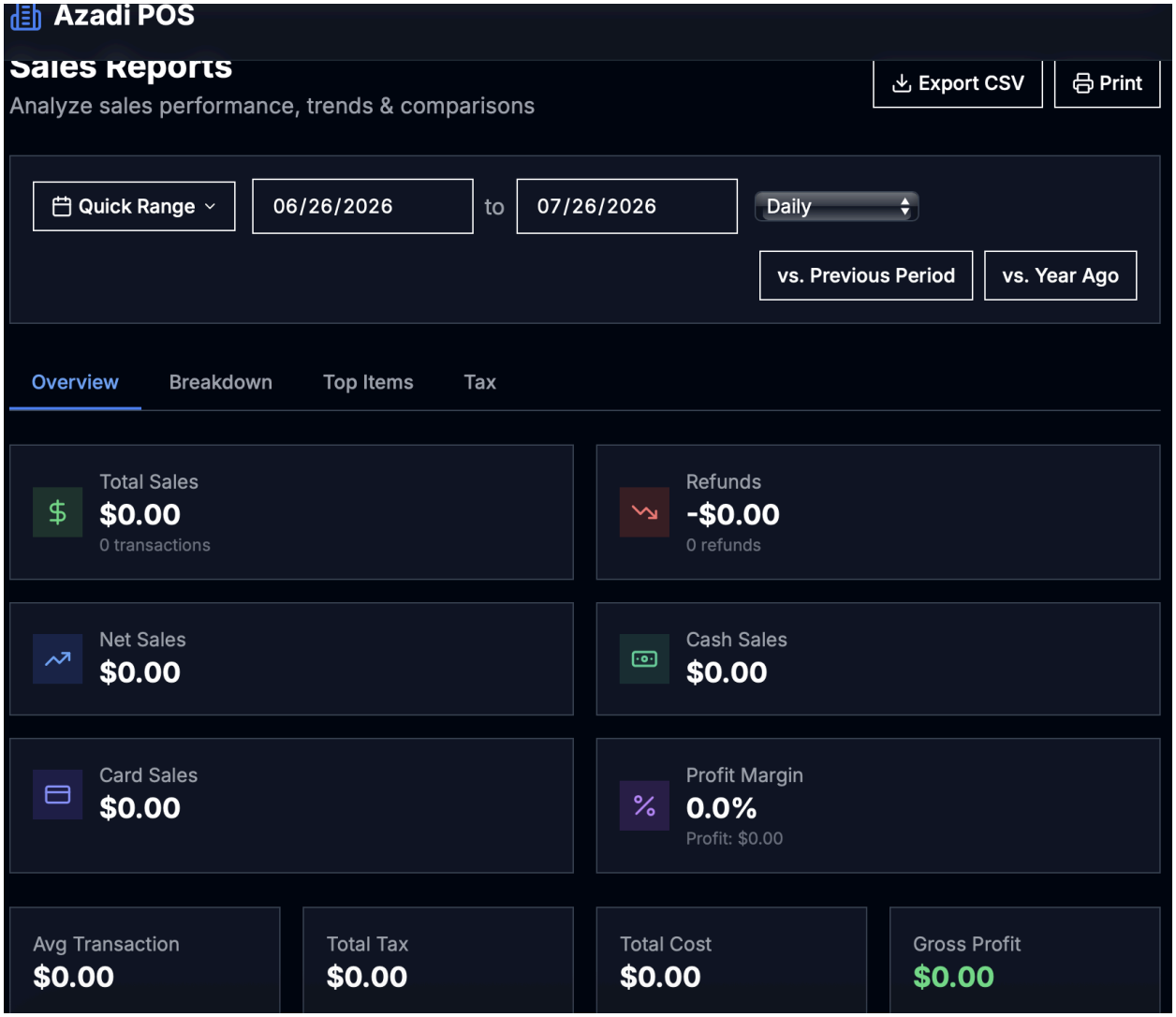


The reorder tile, as mentioned earlier, will alert the user of all items that have hit their reorder quantity level. It will group them by their selected vendor, so the user has a list of all items that need to be reordered by each vendor separately.

For receiving, we want to select the vendor from which items were received, search the barcode, type the quantity, and type the cost. Quantity would add to the quantity on hand in inventory, and price will update in the inventory tab as well in the event of a change. We should be able to add notes if possible. In the event that an item is not in the inventory that is received, we should be prompted with an option to add it to inventory, which would open the same popup as would be opened in the inventory tab.

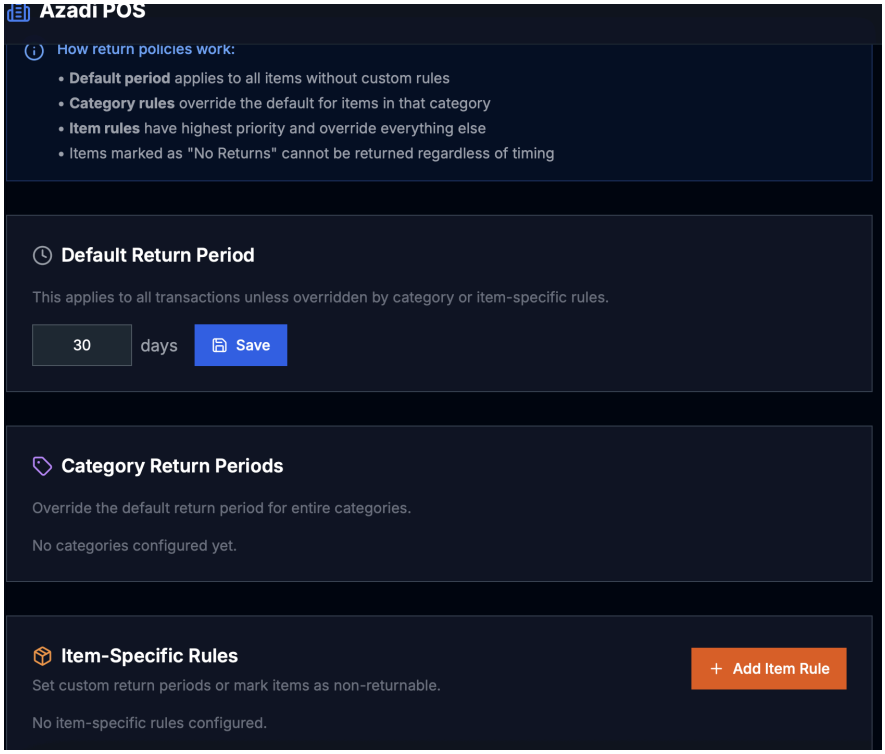
The payouts tab should simply allow the user to log vendor, amount, and any additional notes. This will be used for accounting purposes.

Sales reports allow the user to get a company snapshot with many filter types to view and analyze data, as well as for compliance/filing purposes.

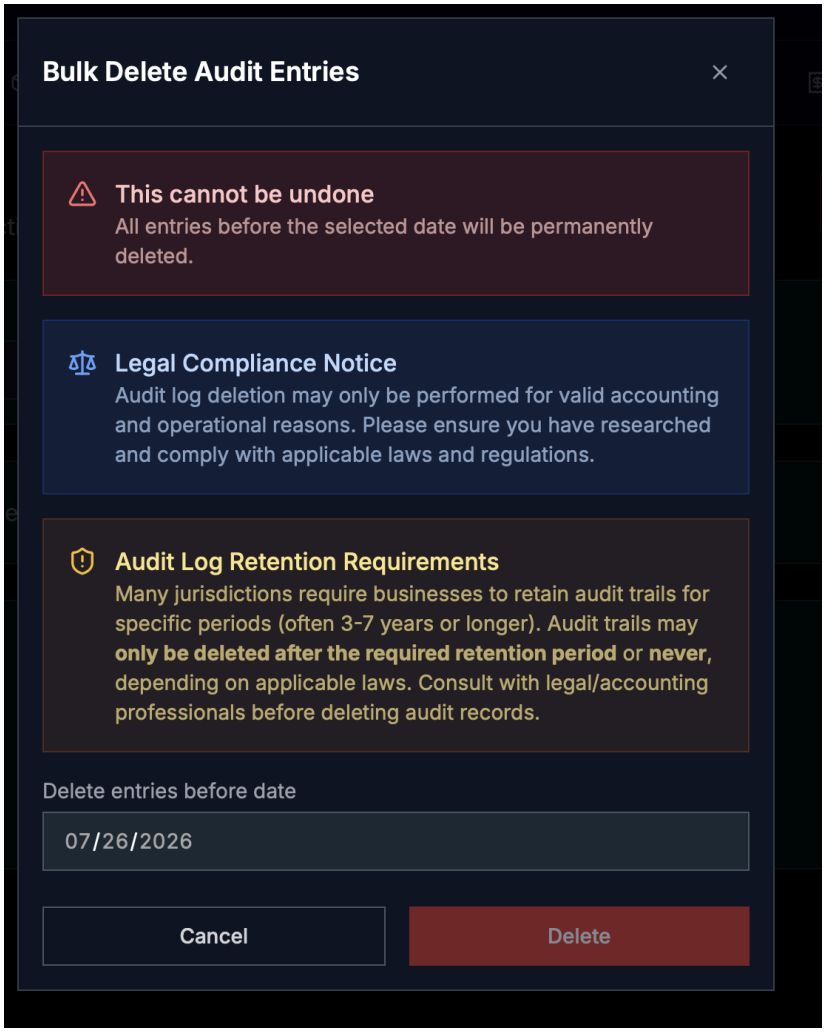


This is a UI that I have come up with that contains all of what I am looking for. An ability to filter the data by a date range, compare the data vs previous periods (though with more flexibility than the two options shown), and payouts should also reflect here somewhere though not shown in the screenshot. Top items and tax are also important tabs for us for analysis and legal reasons.

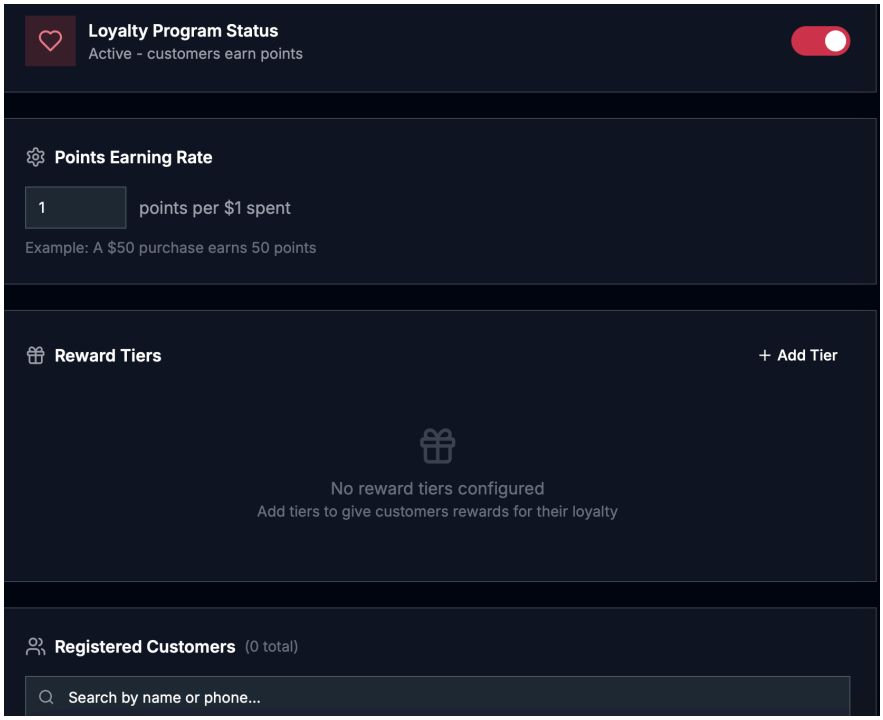
The following picture is a UI I created for setting policies like return periods, and it is perfect for our needs:



An audit trail is a required financial data retention feature for legal reasons. All deleted transactions from the transactions tab should appear here as a history of actions. Again, filters for date ranges and action type (deletes, voids, etc.) are helpful. Before deleting a transaction in the transactions tab, there should be a pop up requiring acknowledgement of various rules. Deleting an audit trail log should do the same. We request the deletion capability to manage storage once the legally required period of time for retaining data has elapsed.



The loyalty program is just a feature to track points earned by phone number. The employee at the terminal should be able to add phone numbers with a name attached in order for that customer to have tracked spending history and earn points.



Redemption types should be able to be configured by the user. We can configure it as dollar amount off, percentage off, with or without minimum purchase required, and name the reward, as well as how many points it will cost to redeem the rewards.



We would like for the ability to generate store only gift cards in-house. We would like to be able to create the unique code, as well as how much is on it. The first time it is scanned and sold will be when it becomes active. The home page of the gift cards section should show us all outstanding cards and their balance. This is a NOT a high priority feature.

Price optimization just allows us to view on a per item basis the sales data at various price points for an item. It allows us to find the sweet spot of qty sold, profit, and price point. This is also NOT a high priority feature.

Lastly, settings allows us to configure our store name, address, and phone number as they would appear on the receipt. We would also like to configure receipt design through uploading a logo image, and header and footer.

The POS terminal side home page should ask for a barcode to log in. This will be the barcode tied to a unique code that was generated from the admin computer. This will identify the terminal's session as associated with a particular employee. Logging in will then welcome that employee along with a standard cashier's numpad asking for the opening balance to be entered. As this involves an entry of financial data which is manipulatable, we want the application to request a managers barcode to approve this entry before allowing the system to continue into the main interface.

The main interface should have 5 buttons. New transaction, end of shift, refund, register, and print barcode.

New transaction has a simple search field on the top where barcodes and names can be searched for or a full barcode can be entered via a connected scanner. **Two NO COMPROMISE features in this area: Typing a barcode must show a highlighted (tentative) completion of the barcode based on best matching dropdown results. This will get further refined as more characters are typed. The rapid input of a barcode (from a scanner) must automatically enter the item.** The right side will just show an order summary. A customer can also be added on this page for earning or redeeming points. The cursor should always stay in the barcode entry field so that the cashier can continue scanning. Multiple scans of the same product should increase quantity.

Barcode printing is for deli purposes. For example, a deli worker should be able to weigh meat, and print the barcode to a label printer with the weight written on it.